Coding Challenge: Teacher Reading Assignment Portal

Overview

In this challenge, you'll design and build a **lightweight, end-to-end web application** that allows teachers to assign reading to students and track assignment status.

This exercise is intentionally **high-level and partially incomplete**. We're interested in how you:

- Interpret requirements
- Make architectural and implementation tradeoffs
- Balance quality with time constraints
- Communicate your decisions clearly

You should aim to spend **approximately 4 hours** on this challenge.

Product Concept: Teacher Reading Assignment Portal

A lightweight portal where teachers can assign books to students and track assignment progress.

User Roles

You may choose your own approach to user identity and roles. We **prefer real authentication**, but expect you to keep scope reasonable.

Teacher

- View a list of available books that can be assigned to students
- Create a reading assignment (book + due date) for a student(s)
- View created assignments, their status, and minutes read
- View assignment status for students

Student

- View assigned reading
- Open/view the assigned book
- Track minutes read
- Update assignment status
- See and update current status:
  - Not Started
  - In Progress
  - Completed

Functional Requirements

Books

- A list of books should be available to assign

Assignments

- Teachers can create assignments by selecting:
  - A book
  - A due date
- Assignments track student progress using the defined statuses

Assignment Status

- Students can update their assignment status
- Teachers can view assignment progress

Intentionally Incomplete Requirements

Some details are intentionally left open.

We expect you to:

- Make reasonable assumptions
- Document those assumptions
- Design for clarity and extensibility

There is no single “correct” solution.

Technical Expectations

Tech Stack

- You may use **any language or framework**
- We primarily work in **Java (backend)** and **React (frontend)**, so feel free to use those if comfortable — but this is **not required.**

Architecture

- Implement both **frontend and backend**

AI Tooling

- You may use **AI-assisted tools**(Claude,Cline, etc) during this challenge.

Deployment

- Application should be:
    - The README should ensure we can run the code locally.
    - Deployed to a public URL (e.g., Vercel, Netlify, Render, Fly.io, etc.)

Deliverables

Please provide:

1. **GitHub repository**
    - Clear README with setup instructions
2. **Live deployed URL with credentials.**

3. **Short written explanation** (README is fine) covering:

    - What you implemented
    - Key architectural decisions
    - Tradeoffs and assumptions
    - What you would improve with more time

Time Expectations

We expect this challenge to take **approximately 4 hours**.

Follow-Up Interview

We'll use your submission as the basis for a potential follow-up discussion, including:

- Architectural decisions and tradeoffs
- Code quality and testing approach
- How you'd lead and evolve this system with a team

There are no trick questions, we want to understand **how you think**.